One-way ANOVA

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Variance; Hypothesis testing: null and alternative.
- **Compare with:** T-tests: one-sample, two-sample, paired.

1. Overview

Key point: One-way ANOVA tests whether three or more group means are equal by comparing two variances: how far the group means lie from each other, and how far the values lie from their own group mean.

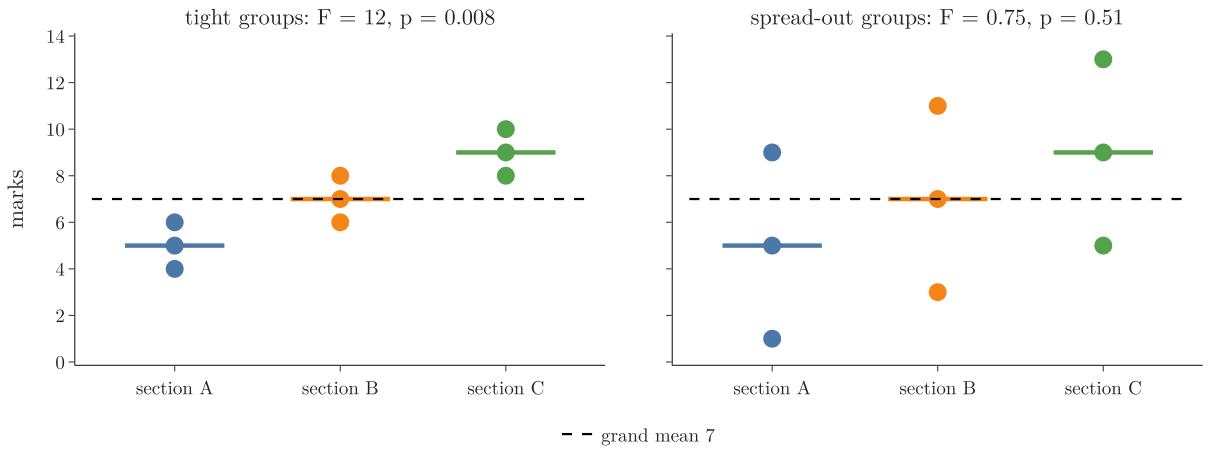


Figure 1: Same group means, different spread: the gaps between means are clear on the left and lost in the noise on the right

ANOVA (**analysis of variance** (G-203)) was named in the tools of inferential statistics as the test that compares the means of several groups. In choosing a hypothesis test ANOVA is the test for one numerical **feature** (G-772) against a categorical feature with three or more categories (a feature is a variable of the data, one column of the data table, such as age or class).

Figure 1 shows the whole idea. Both panels have the same group means, 5, 7 and 9. On the left the values sit close to their group mean, so the gaps between the means stand out. On the right the values are spread widely, and the same gaps could easily be chance. ANOVA turns this comparison into one number, the F statistic.

This Note covers:

- why we do not run a t-test for every pair of groups;

- the split of the total variation into within-group and between-group parts, worked by hand on nine marks;
- the hypotheses;
- the F statistic, the **F distribution** (G-739) and the decision;
- the assumptions, a Titanic case study, and the follow-up question "which groups differ?".

> **Extra:** All of this Note is extra material beyond the "which test when" guide; the guide only names ANOVA.

2. Why not several t-tests?

> **Key point:** Each t-test at $\alpha = 0.05$ has a 5% false-alarm risk; three of them together have about a 14% risk, ten of them about 40%.

A two-sample t-test compares two group means (see the independent two-sample t-test). With three groups A, B and C, we could run three t-tests: A against B, A against C, B against C.

The trouble is the Type I error (see **Type I error** (G-2032)). Each test, run at $\alpha = 0.05$, wrongly rejects a true H_0 5% of the time. Over several tests these risks add up.

1. **In words:** the chance of at least one false alarm is one minus the chance that every test stays quiet.
2. **Formula:** for m independent tests,

$$P(\text{at least one Type I error}) = 1 - (1 - \alpha)^m$$

3. **Example:** three groups need $m = 3$ tests:

$$1 - 0.95^3$$

$$= 1 - 0.857$$

$$= 0.143$$

Five groups need $m = 10$ tests:

$$1 - 0.95^{10} = 0.40$$

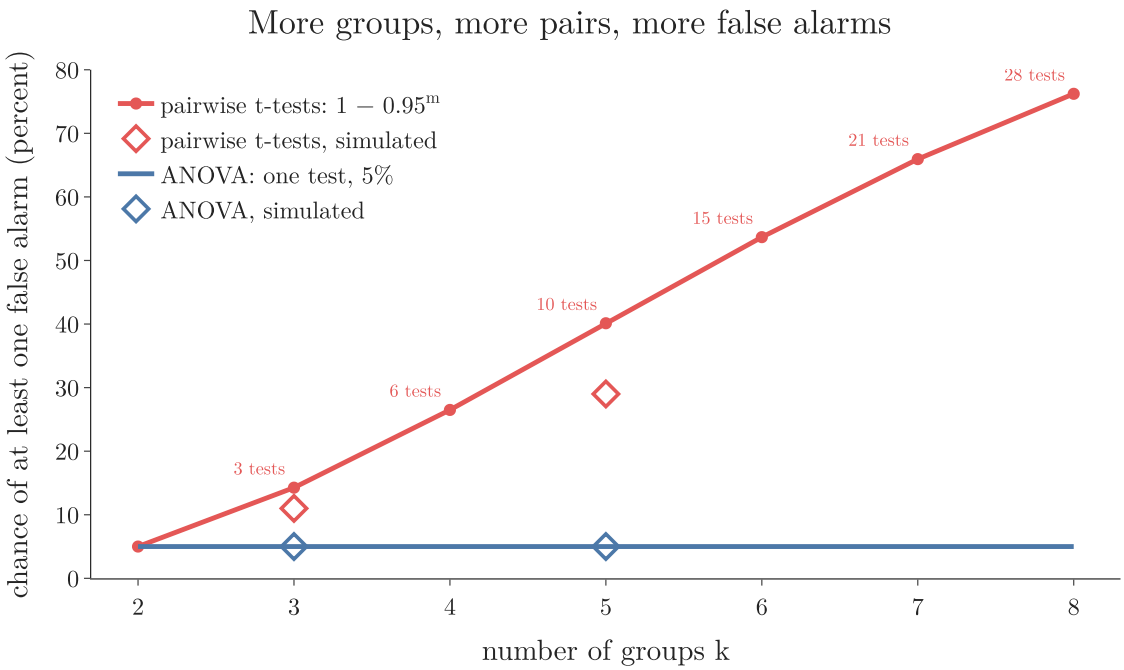


Figure 2: The chance of at least one false alarm: pairwise t-tests (red) against one ANOVA (blue), by formula and by simulation

In Figure 2, follow the red line: every extra group adds more pairs, and the risk climbs while ANOVA stays at 5%. So with five groups whose true means are all equal, pairwise t-tests "find" a difference far more often than 5% of the time. ANOVA asks one question about all groups at once, and its false-alarm rate stays at 5%.

Extra: The formula assumes the tests are independent. Pairwise t-tests are not, since each group appears in several of them, so the real risk is a little lower. The notebook measures the real risk on 2,000 simulated datasets (20 values per group, all true means equal):

Groups	t-tests	At least one false alarm (simulated)	Formula	ANOVA false alarm (simulated)
3	3	11.0%	14.3%	5.0%
5	10	29.0%	40.1%	5.0%

3. Splitting the variation

Key point: Total variation = variation within the groups + variation between the group means: $SST = SSW + SSB$, and the degrees of freedom split the same way.

The data. We carry one small dataset through the whole calculation: the marks (out of 10)

of 3 students in each of $k = 3$ sections of a class.

Section A	Section B	Section C
4, 5, 6	6, 7, 8	8, 9, 10
mean 5	mean 7	mean 9

The **grand mean** (G-866), the mean of all $N = 9$ values, is:

$$63/9 = 7$$

It is also the mean of the three section means, because the sections have equal sizes:

$$(5 + 7 + 9)/3 = 7$$

These are the left-hand groups of Figure 1.

The idea in plain words. The nine marks vary. Some of that variation is noise inside each section: students in the same section still score differently. The rest comes from the sections themselves having different means. **ANOVA** (G-203) measures the two parts and compares them.

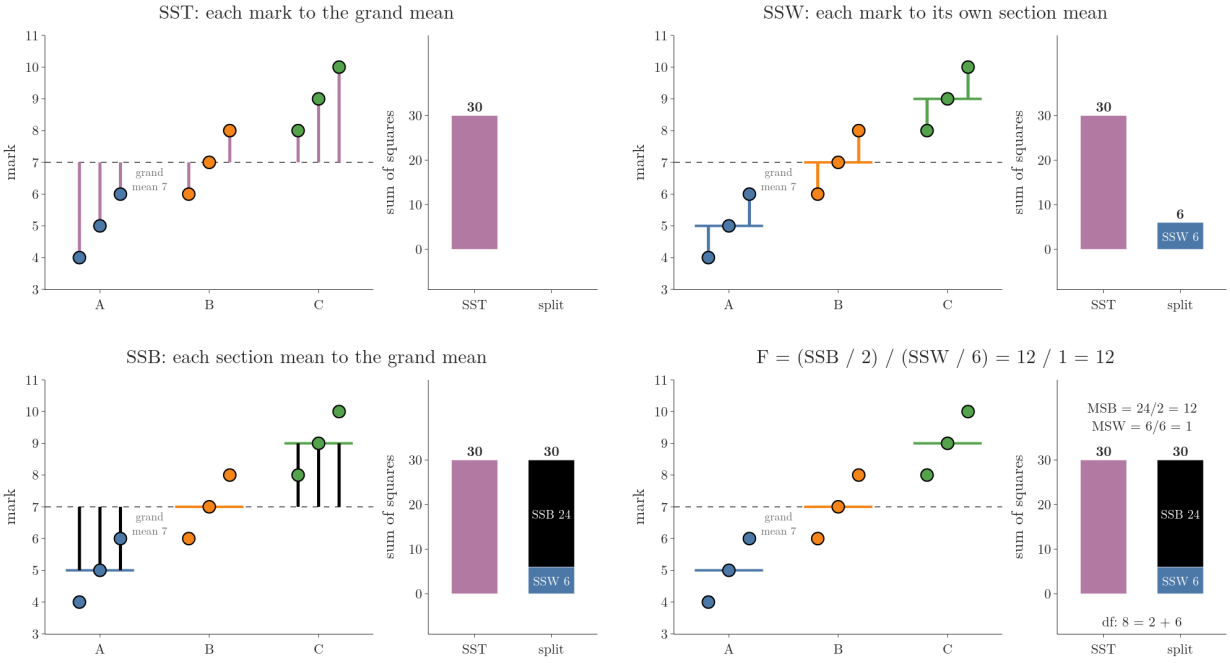


Figure 3: The nine marks as sticks: to the grand mean (SST = 30), to their own section mean (SSW = 6), and from each section mean to the grand mean (SSB = 24); the bars stack $30 = 24 + 6$. Idea after Khan Academy, "ANOVA 2: Calculating SSW and SSB"

Variance is built from squared distances from a mean (see variance). ANOVA works with the **sums of squares** (G-1914), the squared distances added up before dividing. Figure 3 draws each distance as a stick; watch the sticks change as the reference line moves from the grand mean to the section means.

3.1 Total sum of squares

> **Key point:** SST measures how far all values lie from the **grand mean** (G-866).

1. **In words:** square each value's distance from the grand mean and add over all values.
2. **Formula:**
$$SST = \sum_{\text{all values}} (x - \bar{x})^2$$
where $\bar{x}$ is the grand mean and $\sum$ means "add up" (here over all 9 values).
3. **Example:** distances from 7 are $-3, -2, -1$; $-1, 0, 1$; $1, 2, 3$:
$$SST = (9 + 4 + 1) + (1 + 0 + 1)$$
$$+ (1 + 4 + 9)$$
$$SST = 14 + 2 + 14 = 30$$
4. **Degrees of freedom:** $N - 1 = 8$. Once we know the grand mean and any 8 of the marks, the ninth is fixed, so only 8 marks carry free information.

3.2 Within-group sum of squares

> **Key point:** SSW measures how far the values lie from their own group mean: the noise inside the groups.

1. **In words:** square each value's distance from its own group mean and add over all values.
2. **Formula:**
$$SSW = \sum_{\text{groups}} \sum_{\text{values in group}} (x - \bar{x}_i)^2$$
where $\bar{x}_i$ is the mean of group i.
3. **Example:** each section gives:
$$1 + 0 + 1 = 2$$
So the three sections together give:
$$SSW = 2 + 2 + 2 = 6$$
4. **Degrees of freedom:**
$$N - k = 9 - 3 = 6$$
Within each section, once we know its mean and 2 of its 3 marks, the third is fixed: 2 free values per section, 3 sections. This is Bessel's correction (dividing by one less than the count) once per group (see the sample variance).

3.3 Between-group sum of squares

> **Key point:** SSB measures how far the group means lie from the **grand mean** (G-866), each counted once per value in its group.

1. **In words:** replace every value by its group mean, then square that mean's distance from the grand mean and add over all values. Equivalently: square each group mean's distance from the grand mean, multiply by the group size, and add over groups.
2. **Formula:**
$$SSB = \sum_{\text{groups}} n_i(\bar{x}_i - \bar{x})^2$$
where n_i is the size of group i.
3. **Example:**
$$SSB = 3(5-7)^2 + 3(7-7)^2 + 3(9-7)^2$$
$$SSB = 12 + 0 + 12$$
$$SSB = 24$$
4. **Degrees of freedom:** $k - 1 = 2$. Once we know the grand mean and 2 of the 3 section means, the third is fixed.

3.4 The split

$$SST = SSB + SSW$$
$$30 = 24 + 6$$

The degrees of freedom split the same way:

$$8 = 2 + 6$$

Most of the variation here lies between the sections. Figure 4 summarises the split and the steps from there to F.

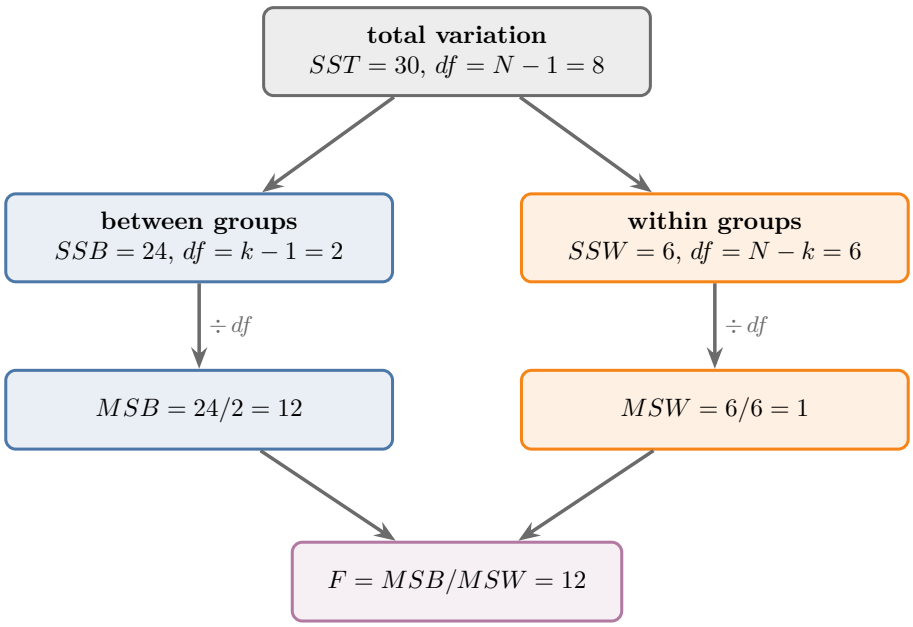


Figure 4: The variation split into its two parts, and the steps from there to F

4. The hypotheses

> **Key point:** H_0: all group means are equal; H_1: at least one group mean differs from the others.

The sections' sample means differ: 5, 7 and 9. Is that a real difference between the sections, or just the luck of which students we happened to look at? The question is about the **population** means, the means over all students who could be taught in each section.

With k groups:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

$$H_1 : \text{at least one } \mu_i \text{ differs}$$

H_1 is not "all means differ". Rejecting H_0 tells us that some difference exists, not which groups differ; section 8 answers that. With three or more groups many patterns fit H_1 (for example "A differs, B and C are equal"), so rejecting H_0 points to "some difference", not to one particular alternative.

"One-way" means one categorical **feature** (G-772) defines the groups: here, the section.

5. The F statistic

> **Key point:** $F = MSB/MSW$: the between-group variance divided by the within-group variance; a large F means the group means differ by more than the noise explains.

5.1 Mean squares

> **Key point:** Dividing each **sum of squares** (G-1914) by its degrees of freedom turns it into a variance, a **mean square** (G-1200).

We use the degrees of freedom of section 3: $k - 1 = 2$ between groups and $N - k = 6$ within groups.

The idea. If H_0 is true, the group means differ only through noise, so the between-group variance and the within-group variance measure the same noise and come out about equal. If the between-group variance is much larger, the variation comes mostly from differences between the means, and H_0 looks wrong.

5.2 F

1. **In words:** divide each sum of squares by its degrees of freedom, then divide the between-group **mean square** (G-1200) by the within-group one.
2. **Formula:**
$$MSB = \frac{SSB}{k-1}$$
$$MSW = \frac{SSW}{N-k}$$
$$F = \frac{MSB}{MSW}$$
3. **Example:**
$$MSB = \frac{24}{2} = 12$$
$$MSW = \frac{6}{6} = 1$$
$$F = \frac{12}{1} = 12$$

If H_0 is true, both mean squares estimate the same noise variance, and F is close to 1 (Montgomery §3.3). Here the between-group variance is 12 times the within-group variance.

For the right-hand groups of Figure 1 (1, 5, 9; 3, 7, 11; 5, 9, 13) the means and so $SSB = 24$ are the same, but $SSW = 96$. Then:

$$MSW = 96/6 = 16$$
$$F = 12/16 = 0.75$$

This is no evidence of a difference.

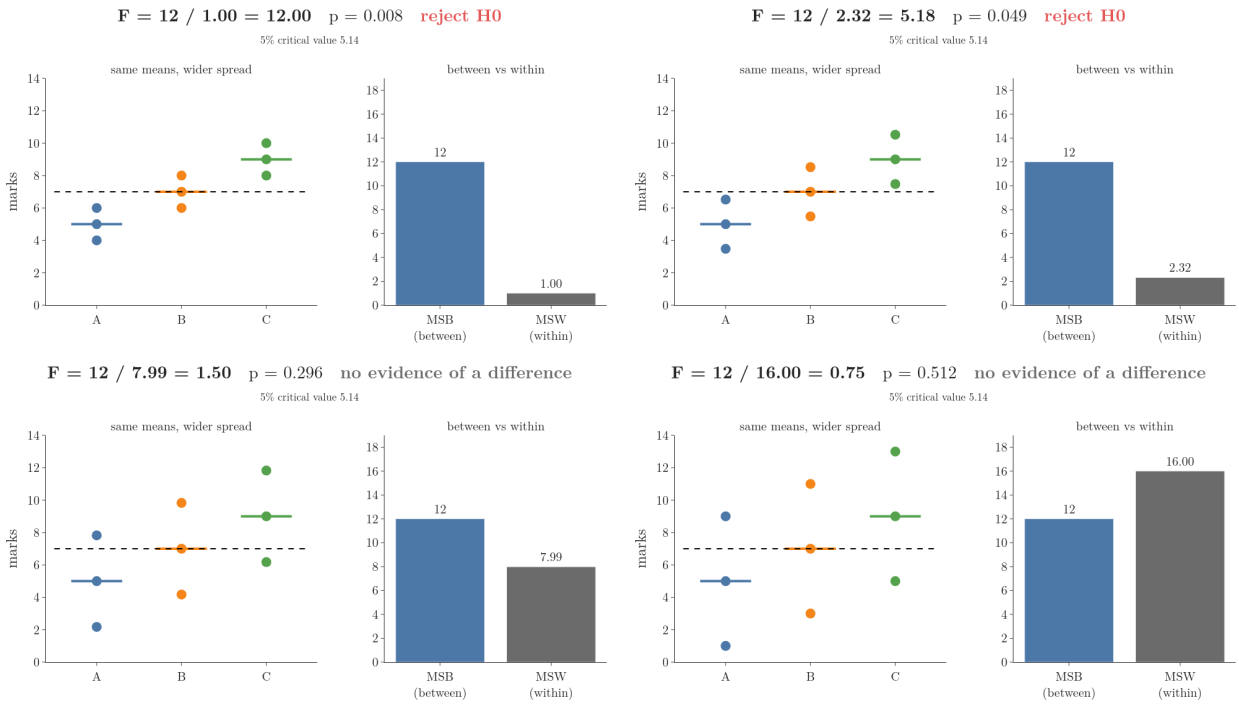


Figure 5: The groups of Figure 1 morphing from tight to spread out: MSB stays 12, MSW grows from 1 to 16, and F falls from 12 to 0.75

In Figure 5, watch the grey bar: only the within-group variance changes, and once F drops below the critical value 5.14 the same group means no longer count as different.

> **Extra:** The results are usually set out as an **ANOVA table** (G-202):

Source	SS	df	MS	F	p
between groups	24	2	12	12	0.008
within groups	6	6	1		
total	30	8			

6. The F distribution and the decision

> **Key point:** Under H_0, F follows an F distribution with $k-1$ and $N-k$ degrees of freedom; the p-value is its right-tail area beyond our F.

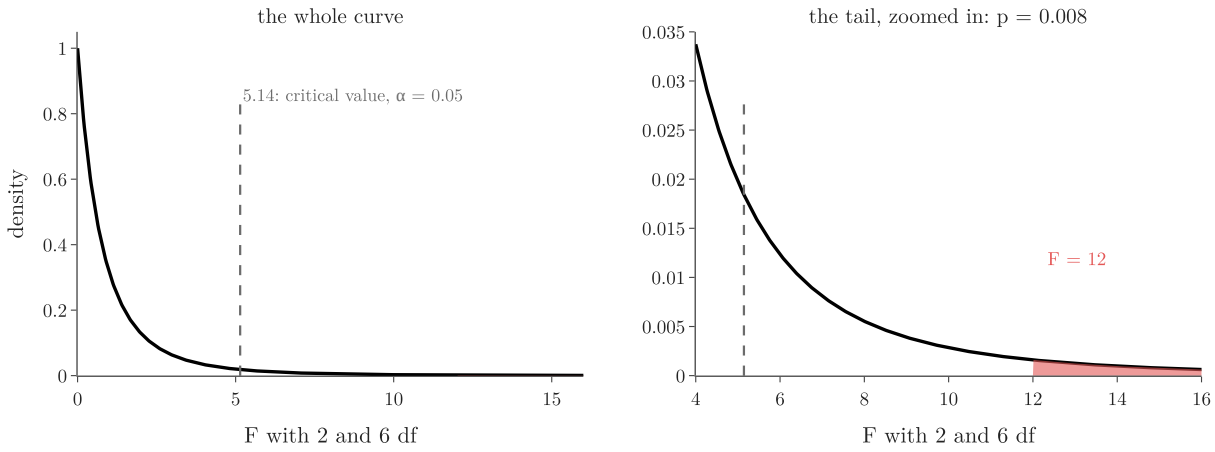


Figure 6: The F distribution with 2 and 6 degrees of freedom: the whole curve (left) and its tail (right)

The **F distribution** (G-739) is the distribution of a ratio of two independent variances (NIST Handbook §1.3.6.6.5). Like the chi-square distribution (see the chi-square distribution), it is never negative and is skewed to the right. The F distribution has two parameters: the degrees of freedom of the top and of the bottom variance.

Only a large F counts against H_0 (group means further apart than noise explains), so the test is always right-tailed. Figure 6 shows the F distribution with 2 and 6 degrees of freedom:

- the 5% critical value is 5.14;
- our $F = 12$ lies far beyond it;
- the tail area beyond 12 is $p = 0.008$.

The critical value depends on α: at $\alpha = 0.10$ it is 3.46, and $F = 12$ is far beyond that too (notebook, section 2b).

Since $0.008 \leq 0.05$, we reject H_0: the three sections do not all have the same mean mark.

Python: `f_oneway` takes one array per group.

```
from scipy import stats

stats.f_oneway([4, 5, 6], [6, 7, 8], [8, 9, 10])
# statistic = 12.0, pvalue = 0.008
stats.f.ppf(0.95, 2, 6)      # 5.14, the critical value
```

Extra: With exactly two groups, **ANOVA** (G-203) and the equal-variance two-sample t-test are the same test: $F = t^2$ and the p-values match. For sections A and B alone, $t^2 = 6$ and $F = 6$.

7. Assumptions

> **Key point:** Independent observations, roughly normal values in each group, and equal variances across groups.

ANOVA (G-203) makes the same assumptions as the two-sample t-test (see its assumptions), for every group:

1. **Independence.** The groups are separate (no one is in two groups), and the observations do not influence each other.
2. **Normality.** The values in each group are roughly normal. With large groups, the central limit theorem makes the group means close to normal anyway (see the conditions of the central limit theorem, which give $n \geq 30$ as the usual guide), so this matters less. Check with a Q-Q plot or the Shapiro-Wilk test (see checking normality).
3. **Equal variances.** The groups have about the same variance; Levene's test checks this, with H_0: "the variances are equal".

When the assumptions fail:

- unequal variances: **Welch's ANOVA** (G-2121) (Welch 1951), or scipy's `stats.alexandergovern`, drops the equal-variance assumption;
- strongly non-normal small groups: the **Kruskal-Wallis test** (G-1019) (`stats.kruskal`; Kruskal and Wallis 1952) compares the groups by ranks instead of means.

8. Case study: age by class on the Titanic

> **Key point:** Mean ages of 38.2, 29.9 and 25.1 years in the three classes give $F = 57.4$ and $p \approx 10^{-23}$; Tukey's test shows that every pair of classes differs.

8.1 The test

> **Key point:** Did the mean age of passengers differ between first, second and third class?

The Kaggle Titanic training file has 714 passengers with a known age.

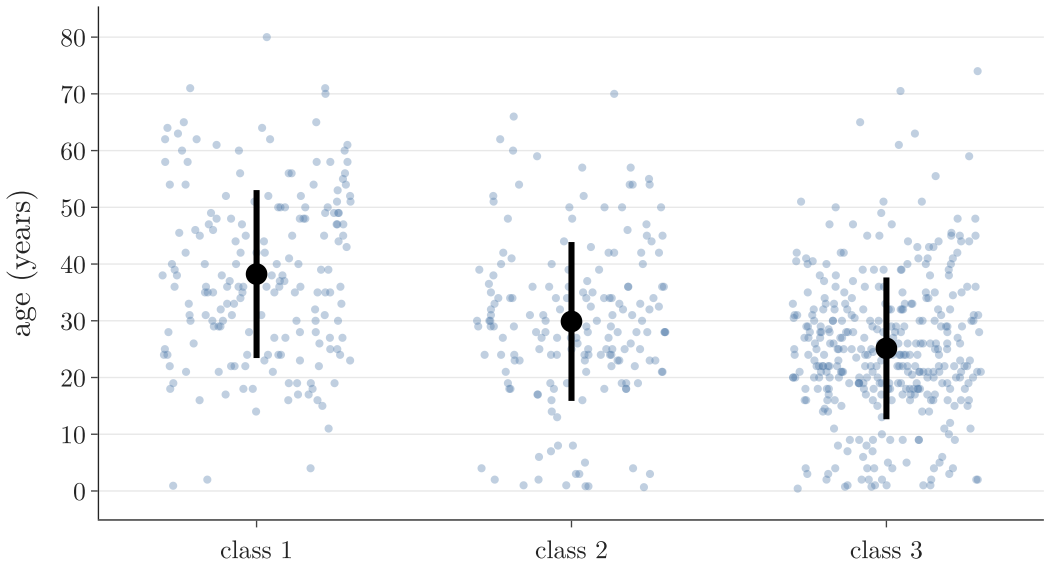


Figure 7: Age of Titanic passengers by class: every passenger as a dot, the class mean and one standard deviation either side in black

Class	Passengers	Mean age	Standard deviation
1	186	38.2	14.8
2	173	29.9	14.0
3	355	25.1	12.5

Figure 7 shows a clear downward step in the means, with much overlap between the classes.

1. **Hypotheses.** $H_0 : \mu_1 = \mu_2 = \mu_3$; H_1: at least one class mean differs.
2. **Significance level.** $\alpha = 0.05$.
3. **Statistic.** The two **sums of squares** (G-1914) are built as in the nine-mark example. SSW adds, over the classes, (passengers $-$ 1) times the variance (the standard deviation squared). Roughly:

$$185 \times 14.8^2 = 40{,}522$$

$$172 \times 14.0^2 = 33{,}712$$

$$354 \times 12.5^2 = 55{,}313$$

$$SSW \approx 40{,}522 + 33{,}712 + 55{,}313$$

$$SSW \approx 129{,}547$$

SSB adds, over the classes, passengers times (class mean $-$ overall mean 29.7) squared. Roughly:

$$186 \times 8.5^2 = 13{,}439$$

$$173 \times 0.2^2 = 7$$

$$355 \times 4.6^2 = 7{,}512$$

$$SSB \approx 13{,}439 + 7 + 7{,}512$$

$$SSB \approx 20{,}958$$

These table values are rounded. The unrounded data give $SSB = 20{,}930$ with 2 df and $SSW = 129{,}527$ with 711 df (the 714 passengers minus 3 classes):

$$MSB = \frac{20{,}930}{2} = 10{,}465$$

$$MSW = \frac{129{,}527}{711} = 182.2$$

$$F = \frac{10{,}465}{182.2} = 57.4$$

4. **P-value.** The p-value is far below 0.05; the 5% critical value is only 3.01:

$$p = 7.5 \times 10^{-24}$$

5. **Decide.** Reject H_0: mean age differed between the classes.

8.2 Checking the assumptions

> **Key point:** Levene's test rejects equal variances ($p = 0.004$), but the tests that drop that assumption agree: the classes differ.

- **Normality:** Shapiro-Wilk rejects normality for classes 2 and 3. With 173 and 355 passengers, the central limit theorem protects the means.
- **Equal variances:** Levene gives $p = 0.004$: the variances (standard deviations 14.8, 14.0, 12.5) differ.

Both checks raise doubts, so we confirm with the tests of section 7. Alexander-Govern gives a p-value of about 4×10^{-21} and Kruskal-Wallis about 1×10^{-21}. The conclusion stands.

8.3 Which classes differ? Tukey's test

> **Key point:** A post-hoc test compares every pair of groups after **ANOVA** (G-203) rejects, while keeping the overall false-alarm risk at 5%.

ANOVA said that at least one mean differs. A **post-hoc test** (G-1534) finds which. The usual choice is **Tukey's HSD** (G-2026) (honestly significant difference), which compares every pair with the overall **Type I error** (G-2032) held at α (Tukey 1949):

Pair	Difference in mean age	p-value
class 1 − class 2	8.4 years	2×10^{-8}
class 1 − class 3	13.1 years	below 10^{-9}
class 2 − class 3	4.7 years	0.0005

Pair	Difference in mean age	p-value

All three pairs differ: first-class passengers were the oldest, third-class the youngest.

Python: ANOVA, the assumption checks and Tukey's test on the Titanic ages.

```
ages = [titanic.loc[titanic["Pclass"] == c, "Age"].dropna()
        for c in (1, 2, 3)]
stats.f_oneway(*ages)           # F = 57.4, p = 7.5e-24
stats.levene(*ages)             # p = 0.004
stats.alexandergovern(*ages)    # p = 4.4e-21
stats.tukey_hsd(*ages)          # every pair
```

9. ANOVA in machine learning

Key point: `f_classif` scores each numerical feature by its **ANOVA** (G-203) F against the class labels, for feature selection.

`SelectKBest` (scores each column against the target and keeps the k best) can score **features** (G-772) (the input variables of a model) with `f_classif` instead of `chi2`. For each numerical feature, `f_classif` computes the one-way ANOVA F with the classes of the **target** (G-1949) (the output we predict) as groups (scikit-learn docs, `f_classif`). A feature whose mean differs strongly between the classes gets a large F, and is kept.

10. Summary

	Several t-tests	One-way ANOVA
Question	is this pair different?	do any of the k means differ?
Type I risk, 3 groups	about 14%	5%
Statistic	t per pair	$F = MSB/MSW$
Distribution	t with $n_1 + n_2 - 2$ df	F with $k - 1$ and $N - k$ df
Follow-up	none needed	Tukey's HSD to find which pairs differ

- **ANOVA** (G-203) asks one question about all groups at once, so its false-alarm risk stays at 5% while the risk of pairwise t-tests climbs with every extra pair.
- $SST = SSB + SSW$: total variation splits into variation between and within groups, so the gaps between the means can be weighed against the noise inside the groups.
- F compares the between-group variance with the within-group variance; near 1 under H_0, because both **mean squares** (G-1200) then estimate the same noise, and large when the means differ.
- The test is right-tailed, because only a large F (means further apart than noise explains) counts against H_0; reject H_0 when $p \leq \alpha$.
- Assumptions: independence, normality in each group, equal variances; **Welch's ANOVA** (G-2121) or Kruskal-Wallis when they fail, so check them before trusting the p-value, as the Titanic case does.
- Rejecting H_0 says some mean differs, because H_1 is only "at least one mean differs"; **Tukey's HSD** (G-2026) says which.
- So to test whether three or more group means are equal, run one ANOVA, not a t-test for every pair.

11. Sources

Built from

- Krish Naik, "Tutorial 32- All About P Value,T test,Chi Square Test, Anova Test and When to Use What?", YouTube, https://www.youtube.com/watch?v=YrhlQB3mQFI. When to use ANOVA (section 1).
- Khan Academy, "ANOVA 1: Calculating SST (total sum of squares)", YouTube, https://www.youtube.com/watch?v=EFdlFoHI_0I. SST and its degrees of freedom (section 3.1).
- Khan Academy, "ANOVA 2: Calculating SSW and SSB (total sum of squares within and between)", YouTube, https://www.youtube.com/watch?v=j9ZPMlVHJVs. SSW, SSB, their degrees of freedom and the split (sections 3.2 to 3.4).
- Khan Academy, "ANOVA 3: Hypothesis test with F-statistic", YouTube, https://www.youtube.com/watch?v=Xg8_iSkJpAE. The hypotheses, F and the decision (sections 4 to 6).

Other references

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12. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
ANOVA (G-203)	A test of whether several groups have the same mean (analysis of variance).
F distribution (G-739)	The shape that a ratio of two independent spreads (variances) follows: a right-skewed continuous distribution with two degrees-of-freedom parameters. ANOVA compares its F statistic with this distribution to get a p-value.
Grand mean (G-866)	The mean of all values from all groups pooled together; ANOVA measures how far the values and the group means lie from it to split the total variation.
Sum of squares (SST, SSB, SSW) (G-1914)	The squared distances from a mean added up, before dividing: total (SST), between-group (SSB) and within-group (SSW), with $SST = SSB + SSW$; ANOVA compares the between-group and within-group parts to test whether group means differ.
Mean square (MSB, MSW) (G-1200)	A sum of squared distances divided by its degrees of freedom, which turns it into a variance (a measure of spread). ANOVA divides the between-group mean square (MSB) by the within-group one (MSW) to get the F statistic.

Term	Meaning
ANOVA table (G-202)	The standard table that sets out an ANOVA's results: for each source of variation (between groups, within groups) it lists the sum of squares, degrees of freedom, mean square, the F statistic and the p-value.
Welch's ANOVA (G-2121)	A version of ANOVA that works even when the groups' spreads differ (it does not assume equal variances).
Kruskal-Wallis test (G-1019)	A test of whether several groups differ that works on the ranks of the values, so it does not assume normality; an alternative to one-way ANOVA.
Post-hoc test (G-1534)	A test run after ANOVA rejects, to find which groups differ.
Tukey's HSD (G-2026)	A follow-up test (post-hoc test) that compares every pair of groups while keeping the overall chance of a false alarm (Type I error) at α.
F statistic (G-740)	One number that compares how far apart the group means are with how spread out the values are inside each group: the variance between the group means divided by the variance within the groups, MSB/MSW. One-way ANOVA computes it; a large F means the group means differ more than the spread inside the groups would explain.
Familywise error rate (G-752)	The chance of at least one false alarm (Type I error) when running several tests: $1-(1-\alpha)^m$ for m tests, each at significance level α.
One-way ANOVA (G-1389)	A test of whether three or more group means are equal, with the groups defined by one categorical column.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Type I error (false positive) (G-2032)	A false alarm: the test declares an effect when there is none (rejecting H_0 when it is actually true); its probability is α.
Target (G-1949)	The output column we predict, such as the class; also called the label.